

# Rajalakshmi Engineering College

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

### Section 1 : Coding

#### 1. Problem Statement

In the faraway land of Arithmetica, there exists an ancient calculator that can only perform bitwise operations. The calculator is locked with a secret code that only works when the number is modified using a special operation called right shifting.

The ruler of Arithmetica, King Thales, needs your help to unlock the calculator. The lock on the calculator is encoded with a number, and the calculator will only open if you apply a right shift by 2 on the number. Your task is to help King Thales determine the magic number that will unlock the ancient calculator.

#### ***Input Format***

The first line of input represents an integer.

### **Output Format**

The output should display the right-shifted value by 2 bits.

Refer to the sample output for formatting specifications.

### **Sample Test Case**

Input: 16

Output: 4

### **Answer**

```
// You are using Java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        System.out.println(n >> 2);
        scanner.close();
    }
}
```

**Status :** Correct

**Marks : 10/10**

## **2. Problem Statement**

In a logistics company, each delivery pack contains a specific number of items, and the priority customer receives double the amount. Write a program to determine the total number of delivery packs required for the operation, considering the number of items per pack and the number of customers given as input by the user.

### **Example**

Input:

Number of items per pack = 96

Number of customers = 8

Output:

10

Explanation:

Given the number of items per pack = 96 and the number of customers = 8, the calculations are as follows:

Total number of items needed = number of items per pack \* number of customers =  $96 * 8 = 768$ . Priority customer's share = double the amount of items per pack =  $2 * 96 = 192$ . Total items with the priority customer = total items needed + priority share =  $768 + 192 = 960$ . Number of packs needed =  $(960 + 96 - 1) / 96 = 10.98$ . Since we cannot have a fraction of a pack, the output is 10.

#### ***Input Format***

The input consists of two space-separated integers N and C, representing the number of items per pack and the number of customers.

#### ***Output Format***

The output displays an integer, representing the total number of delivery packs required for the operation.

Refer to the sample output for formatting specifications.

#### ***Sample Test Case***

Input: 1 1

Output: 3

#### ***Answer***

```
// You are using Java
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```
int n = scanner.nextInt();
int c = scanner.nextInt();
int totalItems = n * c + 2 * n;
int packs = (totalItems + n - 1) / n;
System.out.println(packs);
scanner.close();
}
}
```

**Status :** Correct

**Marks :** 10/10

### 3. PROBLEM STATEMENT:

Julia a mathematician expert is given two integers to find if the second integer is above the average of the first and second integer. Write a program that achieves this using the ternary operator.

#### ***Input Format***

The first line of input represents the first integer.

The second line of input represents the second integer.

#### ***Output Format***

The output should be displayed as "Below Average" or "Above Average"

REFER THE SAMPLE TESTCASES FOR THE FORMAT SPECIFICATIONS.

#### ***Sample Test Case***

Input: 1

1

Output: Below Average

#### ***Answer***

```
// You are using Java
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        int a = scanner.nextInt();  
        int b = scanner.nextInt();  
        double avg = (a + b) / 2.0;  
        System.out.println(b > avg ? "Above Average" : "Below Average");  
        scanner.close();  
    }  
}
```

**Status :** Correct

**Marks :** 10/10

#### 4. Problem Statement:

"Write a program that helps identify the type of a triangle based on the lengths of its three sides. The program prompts the user to input the lengths of sides 'a', 'b', and 'c', and then it classifies the triangle as 'Equilateral' if all sides are equal, 'Isosceles' if two sides are equal, or 'Scalene' if all sides are different. Can you provide the Java code for this task?"

#### **Input Format**

The first line of the input is an integer 'a' representing the length of side 'a.'

The second line of the input is an integer 'b' representing the length of side 'b.'

The third line of the input is an integer 'c' representing the length of side 'c.'

#### **Output Format**

The program outputs a single line that specifies the type of the triangle: "Equilateral," "Isosceles," or "Scalene."

#### **Sample Test Case**

Input: 3

4

5

Output: The triangle is Scalene

**Answer**

```
// You are using Java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int a = scanner.nextInt();
        int b = scanner.nextInt();
        int c = scanner.nextInt();

        if (a == b && b == c) {
            System.out.println("The triangle is Equilateral");
        } else if (a == b || b == c || a == c) {
            System.out.println("The triangle is Isosceles");
        } else {
            System.out.println("The triangle is Scalene");
        }

        scanner.close();
    }
}
```

**Status :** Correct

**Marks :** 10/10